

# Tunisia, audited — the engineer’s ledger

the energy balance, water chemistry, acid-stream assays, lift floors, and every gate  
— everything a technical team verifies before FID.

Provenance law: MEASURED = cited anchor · DERIVED = arithmetic on measured values · ASSUMED = declared default. commercially-validated = 0 — no buyer has signed anything; published as zero on purpose. Generated 2026-09-09 by nations\_audit.py / make\_nations\_pdfs.py — the same code that renders the page.

## 1. The energy ledger, leg by leg

Recovery ledger recomputed: 33.9 GWh/yr recovered + 20 GWh/yr dispatch value. The printed 82 GWh did not sum from its own legs ( $7.2 + 30 + 20 = 57.2$ ; 24.8 unexplained) — corrected, with the arithmetic, on the page.

| LEG              | FORMULA                                                                                                                | VALUE                            |
|------------------|------------------------------------------------------------------------------------------------------------------------|----------------------------------|
| Pump-as-turbines | $9.81 \times (70\% \times 150,000/86,400) \text{ m}^3/\text{s} \times 50 \text{ m} \times 75\% \times 8,760 \text{ h}$ | 3.9 GWh/yr                       |
| Heat recovery    | $0.6 \text{ GJ/t} \times 2 \text{ Mt PG} \rightarrow 333 \text{ GWh thermal} \times 9\%$                               | 30 GWh/yr                        |
| Flexibility      | desal + storage follow the solar curve                                                                                 | 20 GWh/yr (dispatch, not joules) |
| ERD              | inside the 4 kWh/m <sup>3</sup> SWRO benchmark (~35-40% of RO energy)                                                  | 0 added — no double count        |

## 2. Water — the numbers an EPC checks first

| PARAMETER             | VALUE                                                                 | ANCHOR                                                 |
|-----------------------|-----------------------------------------------------------------------|--------------------------------------------------------|
| SWRO specific energy  | 4 kWh/m <sup>3</sup> (benchmark incl. ERD)                            | Rw3 (mega-plant class)                                 |
| Lift physics floor    | 1.09 kWh/m <sup>3</sup> per 400 m at 100% eff; 1.45 at 75%            | Rw4 — the physics that decides where water may travel  |
| Delivered inland cost | \$1.10-1.50/m <sup>3</sup> vs irrigation tariffs \$0.04-0.07          | Rw6/Rw7 — inland water runs on brackish, not long-haul |
| SWRO CAPEX            | \$800-1,200 per m <sup>3</sup> /day (mega-plant)                      | Rw2                                                    |
| Gate                  | blended $\geq \$0.68/\text{m}^3$ , $\geq 60\%$ pre-sold, brine permit | computed (DSCR)                                        |

## 3. Materials — the acid stream

| PARAMETER               | VALUE                       | ANCHOR |
|-------------------------|-----------------------------|--------|
| REE content, Gafsa rock | 322 mg/kg avg, peaks >1,700 | Rm1    |

| PARAMETER           | VALUE                                                       | ANCHOR                   |
|---------------------|-------------------------------------------------------------|--------------------------|
| PG partition of REE | 60-85%                                                      | Rm2                      |
| Acid-stream U       | 85-90% of rock-U partitions to the wet-process acid         | Ru1 (Freeport precedent) |
| U recovery (proven) | >95% DEHPA/TOPO, 14M lbs over 1978-99                       | Ru1                      |
| Radiology           | Ra-226 214-970 Bq/kg in PG; exemption threshold 1,000 Bq/kg | Rm11/Rm12                |
| Phase 0 assays      | 12 stream points: U, REE, Ra-226, Po-210, full metals       | itemized in the audit    |

## 4. The dependencies, by status

| DEPENDENCY                         | STATUS                                                                                                  |
|------------------------------------|---------------------------------------------------------------------------------------------------------|
| Desal brine permit                 | <b>in design</b> — Gabes outfall / diffuser; permit required before SWRO FID                            |
| PG chemistry (Ra-226 partitioning) | <b>Phase 0 assay</b> — flowsheet keeps radium in gypsum (Freeport precedent); re-verify on Gafsa stream |
| Uranium handling                   | <b>gate</b> — CNSTN licensing, NORM transport, export controls                                          |
| REE separation capex               | <b>later phase</b> — +\$244M solvent-extraction plant beyond the ~\$100M circuit                        |
| Solar bankability                  | <b>gate</b> — DSCR ~0.9-1.0 at the record-low tariff; bankable region or grant support required         |
| Grid connection                    | <b>later phase</b> — 3 GW connection + 400 kV upgrades ~\$450M                                          |
| Compute accelerators               | <b>gate</b> — export-control licenses; colocation until an anchor tenant is licensed                    |
| Truffle yield                      | <b>gate</b> — 376 kg/ha is the 20-yr Murcia average; gate = 3-season record >=200 kg/ha                 |
| Export prices                      | <b>exposure</b> — U spot \$80-95/lb; truffle EUR20-40/kg — no contracted offtake yet                    |
| Financing                          | <b>uncommitted</b> — DFI channel modeled for solar/desal; nothing committed                             |

## The water stack — line-by-line, with each credit's failure mode

The \$0.347/m<sup>3</sup> audit: the base is measured anchors; each credit is a loop-internal yield with a named failure mode; the degraded cascade still clears the \$0.68 gate.

| LINE                                                                         | \$/M <sup>3</sup> | PROVENANCE                      | WHAT MAKES IT FAIL                                                                      |
|------------------------------------------------------------------------------|-------------------|---------------------------------|-----------------------------------------------------------------------------------------|
| Base — solar energy 4 kWh/m <sup>3</sup> at the loop's own PPA + capex + O&M | \$0.567           | MEASURED anchors (Rc13/Rw2/Rw3) | holds — if even the PPA lapses, grid power takes the base to \$0.88, still under SONEDE |

| LINE                                                                                                                                                                                                                                                       | \$/M <sup>3</sup>            | PROVENANCE                                                                                | WHAT MAKES IT FAIL                                                                                                                                                                                                                                                                                                                                            |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|-------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DC reject heat → RO feed preheat — one seawater pipe cools the servers, then arrives warm at the membranes (SEC −2.5%/°C)                                                                                                                                  | −\$0.002 net                 | DERIVED — Leg 03 + Leg 01, same pipework                                                  | the trade, stated: warmer feed saves ~\$0.007/m <sup>3</sup> of energy but accelerates biofouling + polyamide hydrolysis (~\$0.005/m <sup>3</sup> membrane debit) — the net is near zero, so the coupling's real justification is the DC's free cooling, not the water credit; lifetime extension comes from pretreatment, thermal stability and flux control |
| Kiln/exotherm preheat — the acid plant and board kilns warm the same feed                                                                                                                                                                                  | −\$0.000                     | DERIVED — Leg 04/05 waste heat                                                            | negligible by design — carried at zero margin                                                                                                                                                                                                                                                                                                                 |
| Brine valorization — NaCl by solar-crystallization ponds (free sun); Mg(OH)&sb2; by lime precipitation from the loop's own kilns (134 t/yr lime, 0.14 GWh/yr — no evaporation), 100 kt to the board's fire retardant + 40 kt surplus sold at the park gate | −\$0.320                     | DERIVED — process specified, CARMEn-anchored cost (Pg-anchor: 2025 techno-economic study) | if the Mg(OH)&sb2; of-take is not signed by FID, the credit degrades to −\$0.03 (disposal avoided only) — the biggest single sensitivity                                                                                                                                                                                                                      |
| Shared civil works — one intake, one outfall, one grid connection serves RO + DC + metals in the same park                                                                                                                                                 | −\$0.032                     | DERIVED — 12% capex sharing                                                               | if the park phases one module at a time, halve it to −\$0.016                                                                                                                                                                                                                                                                                                 |
| Dispatch value — the desal's own flexibility follows the loop's own solar curve                                                                                                                                                                            | −\$0.015                     | DERIVED — 20 GWh shifted                                                                  | fails to zero if the STEG contract does not reward flexibility — contractual, not hoped                                                                                                                                                                                                                                                                       |
| ESG financing delta — the health leg's public KPIs reprice the loop's own debt (100–150bp)                                                                                                                                                                 | −\$0.026                     | DERIVED — nations_v3                                                                      | fails to zero if lenders do not accept the KPI track record — bankable-region fallback keeps the module alive                                                                                                                                                                                                                                                 |
| Brine pump-as-turbine — the desal's own residual pressure                                                                                                                                                                                                  | −\$0.002                     | DERIVED — 2.5 bar × 75%                                                                   | fails to zero — immaterial                                                                                                                                                                                                                                                                                                                                    |
| <b>Integrated</b>                                                                                                                                                                                                                                          | <b>\$0.169/m<sup>3</sup></b> | <b>14% of SONEDE's \$1.00-1.50 (Pg16)</b>                                                 | <b>the degraded cascade: if every soft credit fails, water still lands at \$0.521/m<sup>3</sup> — under SONEDE's own \$1.00–1.50, and the \$0.68 gate still clears with margin. The credits are the margin, not the plan.</b>                                                                                                                                 |

## The people's energy is electrons, not hydrogen — and the H&sb2; lives inside the loop

The loop's H&sb2; (2.63 kt/yr) is an industrial molecule: per kWh of heat, green H&sb2; at \$0.21 is 7× the loop's \$0.031/kWh electron; blending caps at 5-20% by volume (Pg20/Pg21/Pg22); the whole H&sb2; volume equals 1.8% of grid-gas demand. The molecule goes where only a molecule works: ammonia for the GCT chain, inside the loop. the electrolyzer sits on the desal line, and everything it touches stays in the loop:

the water it splits is the SWRO permeate (0.04% of the 150,000 m<sup>3</sup>/d output); its un-split reject stream is clean water that returns to the product line; its oxygen by-product aerates the spirulina ponds and greenhouses; its stack heat (15–20% of input) joins the kiln and DC reject heat on the desal preheat circuit; and the minerals concentrated in its flush stream join the brine valorization line. The electrolyzer is not a consumer of the loop — it is another recovery point inside it.

## The total routing — every stream gets a home

The ponds do *not* drink brine: SWRO brine is 55-70 g/L, past *Arthrospira*'s ~30 g/L ceiling (Pg31); they drink the desal permeate and breathe the electrolyzer's oxygen. The full inventory:

| STREAM                                   | PRODUCED BY                | ITS HOME                                                                                             | THE NUMBER                                   |
|------------------------------------------|----------------------------|------------------------------------------------------------------------------------------------------|----------------------------------------------|
| Solar electrons                          | 500 MW fleet               | desal, metals, DCs, electrolyzer, pumps — routed by the dispatcher                                   | 876 GWh/yr                                   |
| SWRO permeate                            | coastal desal              | city offtake · electrolyzer feed · ponds · greenhouses                                               | 150,000 m <sup>3</sup> /d                    |
| SWRO brine                               | coastal desal              | salt + magnesium valorization, evaporated on the loop's own free heat                                | 122,727 m <sup>3</sup> /d → 58 kt Mg/yr      |
| Brine bittern                            | valorization               | Mg(OH) <sub>2</sub> ; fire retardant → <b>the board's own chemistry</b>                              | 140 kt/yr = 140% of the board's need         |
| Electrolyzer oxygen                      | H <sub>2</sub> ; plant     | spirulina pond + greenhouse aeration                                                                 | 21 kt/yr                                     |
| Electrolyzer hydrogen                    | H <sub>2</sub> ; plant     | ammonia → the GCT fertilizer chain                                                                   | 2.63 kt/yr                                   |
| Electrolyzer reject water / heat / flush | H <sub>2</sub> ; plant     | product line / desal preheat / brine line                                                            | nothing leaves                               |
| DC reject heat                           | compute park               | the same seawater pipe, arriving warm at the membranes                                               | −\$0.007/m <sup>3</sup>                      |
| Kiln + acid exotherm                     | board kilns + metals plant | desal preheat + greenhouses                                                                          | 30 GWh/yr                                    |
| CO <sub>2</sub> ; (kiln + power)         | combustion streams         | mineralized into board (2 Mt PG) · pond carbon                                                       | 540 t CO <sub>2</sub> /yr to the ponds       |
| Phosphogypsum                            | GCT stack                  | fire-rated board — with its retardant from our own brine                                             | 2 Mt/yr                                      |
| Acid stream                              | GCT wet-process acid       | U + heavy REE extraction; raffinate back to fertilizer                                               | 716.5k lbs U <sub>2</sub> O <sub>8</sub> /yr |
| Municipal return flows                   | city mesh                  | 30%+ reuse · pump-as-turbines on the descent                                                         | 3.9 GWh/yr                                   |
| Sludge / digestate                       | wastewater + organics      | biogas for kilns and greenhouses                                                                     | booked option, honestly labeled              |
| Slaughterhouse bones                     | local slaughterhouses      | bone char → the fluoride filters; <b>spent char</b> → <b>phosphate soil amendment for the fields</b> | filter loop closes into fertilizer           |

| STREAM                        | PRODUCED BY           | ITS HOME                                               | THE NUMBER                                   |
|-------------------------------|-----------------------|--------------------------------------------------------|----------------------------------------------|
| Spent pond medium             | spirulina harvest     | nutrient-rich → greenhouse fertigation                 | nothing discarded                            |
| Ammonia fertilizer            | GCT chain             | the terfas fields' host plants + greenhouse soils      | the truffle grows on the loop's own nitrogen |
| Compute services              | colo park             | export — the data product                              | tenant-gated                                 |
| Ra-226 / Po-210               | acid stream radiology | stays in the gypsum/board matrix — the radiology gate  | below the 1,000 Bq/kg exemption              |
| Board offcuts                 | board line            | recycled into new board                                | zero waste                                   |
| Sodium chloride               | brine line            | industrial offtake at the park gate                    | nothing hauled                               |
| Halophytes / Artemia on brine | brine line            | future option, honestly labeled — not a current credit | the only stream awaiting its consumer        |

the ponds drink the desal permeate —  $\sim 0.2$  m<sup>3</sup>/d,  $\sim 2\%$  of the 150,000 m<sup>3</sup>/d output — and breathe the electrolyzer's oxygen; the brine (55–70 g/L, far past the  $\sim 30$  g/L ceiling the organism tolerates) is routed to salt and magnesium valorization instead. Halophytes and Artemia on brine are booked as a future option, honestly labeled — not a current credit.

## The fuel invoice, honestly split — the 500 MW is the grid's medicine, not the whole prescription

of the \$4.895B fuel invoice, the 500 MW fleet displaces only the power-sector gas slice —  $\sim \$0.75$ B/yr (DERIVED from the energy balance) — by \$106-125M/yr, which is 14.1-16.7% of that slice. The rest of the invoice has its own answers, not solar's: the bottles ( $\sim \$0.55$ B of LPG, the bonbonne) leave the market as the people shift to electrons — induction on the loop's own surplus — and the biogas pilot; transport fuels ( $\sim \$2.6$ B) are the honest frontier. The 500 MW is the grid's medicine, not the country's entire prescription.

## The mountains, honestly analyzed

Fog collection is real where fog lives (10-20 L/m<sup>2</sup>/day, Pg33) — Tunisia's fog lives in the already-humid NW; the truffle mountains are the arid interior, and cloud seeding has no proven routine gain (Pg32). What the mountains give the loop: the mountains enter the loop as storage, rainfall and free cooling — and as the land where the truffle belongs (freeing the lowlands that grow today's food). Fog harvesting from pipe meshes stays off the program: the moisture is not there to collect, and the honest scientist does not plant a net under a cloudless sky.

## The AI regulation — five layers, down to the drippers

The dispatcher is a model-predictive controller on a 15-minute receding horizon over the mass balances (Lexicographic: safety > contracts > economics); the twin trains it offline and audits it online, with a Kalman filter for unmeasured states and a published prediction-error KPI; node controllers stay PID-class with hard-wired interlocks; the governance layer never delegates a gate.

| LAYER                                                                           | WHAT IT DOES                                                                                                                                                                                                                                                       | THE MATHEMATICIAN'S NOTE                                                                                                                                                                                                                                                                                 |
|---------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Layer 0 — the instrumented field                                                | every node is a sensor + an actuator: membrane pressure and conductivity, brine concentration, pond pH and temperature, electrolyzer stack voltage, kiln thermocouples, PAT speed, reservoir levels, household-filter telemetry, the grid tie, the weather station | the stream inventory above is the measured state — the mass balances are the constraints the controller must never violate                                                                                                                                                                               |
| Layer 1 — node controllers (edge, millisecond-to-second)                        | local closed loops of PID class: membrane pressure, pond dosing, kiln temperature, pump speed — with hard-wired safety interlocks                                                                                                                                  | functional safety is never on the AI: a controller that fails safe is a relay, not a model                                                                                                                                                                                                               |
| Layer 2 — the dispatcher (model predictive control, 15-minute receding horizon) | a multi-variable MPC: state = solar forecast, reservoir inventory, brine concentration, electrolyzer turndown, DC load, grid price, offtake commitments; decisions = pump flows, setpoints, brine routing, dispatch; forecasts 36–48 h weather, 24 h load          | objective: minimize loop energy cost + degradation + gate-violation penalties, subject to mass balances, storage states and contract floors — lexicographic, never weighted: safety > contracts > economics. Its measured output is the 20 GWh/yr dispatch value and the routing-dependent water credits |
| Layer 3 — the digital twin (simulation, learning, audit)                        | a physics-calibrated model of every balance: trains the dispatcher offline, audits every decision after the fact, simulates before any physical change                                                                                                             | unmeasured states (membrane fouling, pond bloom state) are estimated with a Kalman-class filter from pressure-flow and optical residuals; the twin's prediction error is a published KPI — the AI's accuracy is measured, not claimed                                                                    |
| Layer 4 — the governance layer (the boundary, unchanged)                        | the scoreboard computes and publishes every gate number from Layer 0 telemetry with provenance; the price index; the results framework; anomaly flags to humans                                                                                                    | the AI never sets a gate, never signs, never prices — the falsification ladder is mechanical thresholds, not a learned model. The nervous system is a servant, not a master                                                                                                                              |

| NODE            | SENSES                                                            | CONTROLS                                                                  | TIMESCALE       | LAYER |
|-----------------|-------------------------------------------------------------------|---------------------------------------------------------------------------|-----------------|-------|
| PV field        | irradiance, string current, panel temperature, soiling sensors    | inverter setpoints, curtailment, cleaning scheduling                      | seconds–minutes | L1–L2 |
| Desal trains    | feed pressure, permeate conductivity, ERD pressure, CIP condition | high-pressure pump, ERD bypass, antiscalant dosing, CIP cycles            | seconds–hours   | L1–L2 |
| Membrane health | pressure-flow residuals (fouling inferred by Kalman filter)       | maintenance scheduling, train rotation                                    | days            | L3    |
| Brine line      | concentration, temperature, Mg saturation                         | evaporator feed rate, Mg(OH) <sub>2</sub> precipitation setpoints         | minutes         | L1–L2 |
| Electrolyzer    | stack voltage, current, purity, water feed                        | turndown, ramp rate, O <sub>2</sub> vent routing to ponds, reject routing | seconds         | L1–L2 |
| Board kilns     | temperature profile, feed rate, offcut fraction                   | burner setpoints, retardant dosing, offcut recycle                        | minutes         | L1–L2 |

| NODE                                | SENSES                                                                            | CONTROLS                                                                                                                                                                          | TIMESCALE          | LAYER |
|-------------------------------------|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------|-------|
| Metals plant                        | SX stream assays, raffinate quality                                               | extraction/stripping setpoints, raffinate return to fertilizer                                                                                                                    | minutes–hours      | L1–L2 |
| DC park                             | cooling-loop temperature, PUE, workload mix                                       | seawater flow to racks, batch-workload scheduling to solar peaks                                                                                                                  | minutes            | L2    |
| Water mesh                          | reservoir levels, pump status, pressure heads, PAT speed                          | pump scheduling, zone valves, PAT bypass, the solar-following curve                                                                                                               | 15-minute dispatch | L2    |
| Spirulina ponds                     | pH, salinity, temperature, optical density                                        | aeration timing, CO <sub>2</sub> injection, medium exchange, harvest scheduling                                                                                                   | hours              | L1–L2 |
| Greenhouses                         | EC, pH, humidity, CO <sub>2</sub> , soil moisture                                 | fertigation dosing (spent medium), ventilation, CO <sub>2</sub> enrichment from the kiln stream                                                                                   | minutes            | L1–L2 |
| Establishment drip — to the emitter | per-zone capacitance soil-moisture probes, per-zone flow meters, weather forecast | zone valve actuators per management block, pump pressure, fertigation dosing — emitters stay passive pressure-compensated; clogged drippers detected from per-zone flow residuals | hourly–daily       | L1–L2 |
| Household filters                   | usage telemetry, replacement flags                                                | fleet logistics: replacement routing, spare inventory, the pilot's uptime KPI                                                                                                     | days               | L2–L4 |
| Grid tie                            | import/export price, solar forecast, offtake commitments                          | import vs. dispatch decisions                                                                                                                                                     | 15-minute          | L2    |
| Scoreboard + cohort                 | every KPI stream above, provenance-hashed                                         | gate computation, price index, anomaly flags to humans                                                                                                                            | continuous         | L4    |

**what the AI is honestly worth: the 20 GWh/yr dispatch (~\$0.8M/yr) + the routing-dependent water credits it executes (~\$0.05-0.08/m<sup>3</sup> — DC heat, kiln heat, brine heat, dispatch) + a DERIVED 1–3% park OPEX from multi-variable coupling no manual schedule can follow — and the scoreboard, which is the program's entire credibility.** if the AI is down, the loop does not stop: Layer 1 runs every node in safe, suboptimal mode — the degraded cascade already computed (\$0.521/m<sup>3</sup> water, gate still clear). AI uptime is a KPI, not a dependency.

## The irrigation — to the dripper, honestly

the fields drink at establishment, then live on rain — and the greenhouse loop runs on the pond's own spent nutrients. The irrigation system is designed the way the truffle itself is: drought-proof by architecture, not by subsidy. Establishment: 624 m<sup>3</sup>/ha/yr for the first two years → 4,274 m<sup>3</sup>/d for 2,500 ha, from inland brackish water and reuse; then rainfed.

## Cloud generation — the honest version

our 12 km<sup>2</sup> array absorbs ~156 MW more sunlight than the bare soil (albedo Δ0.05) — a persistent local heat source, plus the ponds' evaporation placed upwind of the fields. Li et al. (Pg34) measured the class of effect at continental scale; at ours it is a designed microclimate, not a rain machine: soil shaded by panels retains moisture, the updraft adds marginal convective activity, and the dew that follows is free water on the

host plants. The honest claim is directional and modest — and it costs nothing extra: the array was being built anyway.

## The mountain portfolio — if mountain agriculture is needed, the version that scales

| CROP / SYSTEM                                          | HECTARES                                                 | WATER                                                                            | VALUE                                                                          | THE SCALE LOGIC                                                                                                     |
|--------------------------------------------------------|----------------------------------------------------------|----------------------------------------------------------------------------------|--------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|
| Terfas belt (arid jebels)                              | 2,500-3,000 ha                                           | rainfed + micro-catchments + dew                                                 | €15-45M/yr at scale (market-capped)                                            | already the design: the drought-proof luxury crop on land nothing else farms                                        |
| Cactus pear (opuntia) belt                             | consolidate + expand the existing ~600k ha national belt | zero irrigation; 300-500 mm survives on 150                                      | fruit at 20-40 t/ha for the basin's price index + cladodes as livestock fodder | the arid mountain staple — Tunisia is already a world producer; the program adds value chains, not water            |
| Medicinal & aromatic plants (rosemary, thyme, oregano) | 5,000-10,000 ha across the Dorsale                       | rainfed; 300-500 kg/ha dry                                                       | ~€8-20M/yr export at €4/kg dry                                                 | the highest value-per-kilo crop that needs no water at all — the mountain cash crop #2                              |
| Agrivoltaic fodder (sulla/alfalfa under the PV field)  | up to 1,250 ha of shaded soil under the 500 MW array     | the panels' shade halves evaporation; no added water beyond the field's own rain | fodder for ~5,000-10,000 head; the flock fertilizes the field                  | the loop's livestock leg: sheep graze solar parks already (Spain/US precedent) — the AI routes the grazing schedule |
| Rainwater harvesting (swales + terracing)              | 2-3% of mountain catchment area                          | each ha of swales captures ~3,000 m <sup>3</sup> /yr at 300 mm rainfall          | \$200-500/ha — an order of magnitude under drip-system capex                   | the mountains' real water system: catch the rain that already falls there, at scale, without pipes                  |

mountain agriculture at scale means putting each crop where its water cost is zero: terfas, cactus, MAPs and agrivoltaic fodder on the mountains' own rain and the loop's own shade — while wheat, vegetables and orchards stay on the lowlands and in the greenhouses where water is abundant. The mountains feed the people what only mountains can grow; the pipes carry water only where gravity and economics both say yes.

## The growroom at landscape scale — climate steering toward bio-recovery

**Measures:** Atmospheric transect: RH / temperature / dew-point stations up the slope at the mist corridor, leaf-wetness sensors, 2-3 sonic anemometers for the anabatic flow; Soil grid: capacitance moisture probes per management zone (the drip grid doubled as the climate grid), soil temperature; The sky: satellite NDVI every 5 days into the twin, rain gauges along the corridor, pond evaporation pans; The park: panel albedo/soiling sensors (the field's reflectivity IS a climate actuator), pond water temperature

**Steers:** The mist corridor: fog nozzles, anabatic hours only, virtual-temperature-positive dosing — 2% of desal flow; The shade: the PV array's own footprint: shaded soil retains moisture; cleaning schedule shifts albedo; The pond surface: evaporation timing follows the wind — the ponds breathe upwind of the fields on purpose; The swales: rainfall harvested at contour, released by zone valves into the host-plant belt

**Objective:** the bio-recovery term in the dispatcher’s objective: maximize dew days + soil-moisture persistence above wilting point + NDVI slope — subject to the energy budget, the thermals remaining buoyant, and the rain-gauge referee. The twin adds a soil-water-balance module (Penman-Monteith evapotranspiration, standard equations, computed in code) so every steering decision has a predicted and a measured outcome.

**The spiral:** the feedback the steering rides: mist → dawn dew → soil moisture → host plants establish → transpiration raises local humidity → more dew — the humidity loop that turns a slope back green, one feedback cycle at a time. The KPIs published: dew days per month, days above wilting point, NDVI slope — bio-recovery, measured like a growroom, audited like everything else.

## The invisible drop — and the effects ledger, direct and side

the drop → microbial pulse → nutrient mineralization → plant-available N and P → host-plant growth → nectar and pollen → the insects: wild bees at 1,000-2,000 flower visits a day, ants engineering the soil, beetles on the wind → the host plants set seed (Helianthemum is insect-pollinated) → the mycorrhizal trade persists → the truffle → the harvest → the price index. Every dew morning the mist buys is 10<sup>8</sup>; cells per square centimeter of leaf waking up — and the corridor's vegetation is the pollinator highway the truffle's own host plants depend on. One invisible drop feeds the microbes that feed the plants that feed the insects that pollinate the plants that feed the truffle that feeds the people.

and the drop re-enlists the army Tunisia lost: the right-of-way vegetation is the refuge for the beneficials — ladybirds (5,000 aphids each in a lifetime), lacewings, hoverfly larvae (400-600 aphids each), and the parasitoid wasps Tunisia already proved at national scale in the date oases (Trichogramma against the date moth). Every predator the corridor hosts is a pesticide that was not bought, not sprayed, and not inhaled by a child — biocontrol is the health ledger's quietest leg.

| EFFECT                                | CLASS  | THE MODEL                                                        | THE INSTRUMENT                         |
|---------------------------------------|--------|------------------------------------------------------------------|----------------------------------------|
| Mist vapor transport                  | DIRECT | Gaussian plume + anabatic advection                              | sonic anemometers, RH transect         |
| Dew deposition on host plants         | DIRECT | surface energy-balance dew model (T <sub>surf</sub> < dew point) | leaf-wetness sensors, dew gauges       |
| Soil moisture gain, corridor zone     | DIRECT | Penman-Monteith water balance                                    | capacitance probe grid                 |
| Seedling establishment survival       | DIRECT | water-stress survival curves                                     | nursery registry, field counts         |
| PV-shade moisture retention           | DIRECT | shade fraction × PET reduction                                   | soil probes under panels               |
| Thermal buoyancy (the constraint)     | DIRECT | virtual-temperature profile                                      | T/RH + wind stations on the slope      |
| Microbial Birch pulse                 | SIDE   | rewetting respiration curves                                     | soil CO <sub>2</sub> ; efflux chambers |
| Beneficial insect recruitment         | SIDE   | habitat-area × predator-prey dynamics                            | sticky traps, acoustic/AI counters     |
| Pollinator visitation → host seed set | SIDE   | pollination-saturation curves                                    | visit counts, seed counts              |

| EFFECT                                  | CLASS                | THE MODEL                                             | THE INSTRUMENT                                     |
|-----------------------------------------|----------------------|-------------------------------------------------------|----------------------------------------------------|
| Precipitation return                    | SIDE — ZERO-credited | cloud microphysics (the least-solved field, Pg32)     | rain-gauge network                                 |
| Pesticide displacement → child exposure | SIDE                 | exposure-response from the cohort                     | urinary pesticide metabolites                      |
| Atmospheric dust & PM scavenging        | SIDE                 | wet-deposition scavenging coefficients (dew/fog wash) | PM sensors along the corridor                      |
| Air quality, basin-wide                 | DIRECT               | emissions inventory → dispersion model                | PM2.5/NOx/SO2 grid + the cohort's respiratory KPIs |

## The land, projected — the Darwinian view of the environment itself

| LAYER OF LIFE    | WITHOUT THE PLAN                                                                                                                  | WITH THE PLAN                                                                                                                                      |
|------------------|-----------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|
| Soil biology     | the seed bank dies with the soil — desertification's positive feedback: less vegetation, less humidity, less dew, less vegetation | the Birch pulse wakes with every dew morning; the microbial city rebuilds; 10 km of hyphae per gram of soil come back to trade                     |
| The insects      | beneficials stay locally extinct; the pesticide treadmill accelerates; pollinators thin, host plants set less seed                | the right-of-way is the refuge: ladybirds, lacewings, hoverflies, Trichogramma's cousins re-recruit; every predator is a pesticide not bought      |
| The dew economy  | bare, crusted ground sheds what little rain falls; the drop dies on arrival                                                       | the mist corridor turns the dawn into a harvest: dew on host plants, soil moisture banked, the spiral compounding                                  |
| The truffle belt | the wild harvest shrinks; the mycorrhizal network starves                                                                         | the belt establishes on the corridor's moisture: 100 ha, then 2,500-3,000 ha, on the land the humidity itself prepared                             |
| The frontier     | desertification marches downhill — the slope's fitness landscape selects for stone                                                | each humidified season pushes the establishment frontier a little further upslope — the landscape's selection flips from bare ground to vegetation |

the Darwinian projection of the whole machine: the same land, two futures — one where the environment keeps selecting for stone, and one where a 2% mist margin re-engineers the fitness landscape so that life out-competes bare ground. The honest bound stands: the corridor transforms a belt, not a mountain range — the wider slopes change only through the slow, compounding spiral and the rain cycle. We plant the drop; the land grows the rest.

## The nature ledger — every class of credit, not just carbon

| SPECIES / ASSET                        | THE HONEST NOTE                                                                                                                 |
|----------------------------------------|---------------------------------------------------------------------------------------------------------------------------------|
| Wild bee community                     | ~700+ species documented in Tunisia; the corridor's nectar belt is their refuge and the truffle host plants' pollination engine |
| Egyptian vulture (globally endangered) | breeds in Tunisia; scavenger of the open steppes the corridor stabilizes                                                        |

| SPECIES / ASSET              | THE HONEST NOTE                                                                                      |
|------------------------------|------------------------------------------------------------------------------------------------------|
| Dorcas gazelle               | endangered; the restored steppe of the south is its historic range                                   |
| Cork oak + orchid flora (NW) | the humid mountains' own endangered flora — the value-chain program that ends over-harvesting        |
| Medicinal & aromatic plants  | wild-rosemary pressure relieved by the cultivated MAP belt — the market saves the wild stock         |
| Addax                        | honestly named: locally extinct in the wild — NOT claimed; restoration is a later, separate question |

the nature ledger: the corridor's ~50 km<sup>2</sup> of restored belt at maturity earns **\$0.0-0.2M/yr** of biodiversity credits at the nascent-market band (\$5-50/ha, DERIVED — booked at the low end; TNFD disclosure framework, Pg37; GBF Target 19, Pg36) and unlocks a **debt-for-nature swap**: \$500M of the state's debt refinanced against measured nature outcomes — ~\$7.5M/yr of coupon savings, the Gabon/Ecuador instrument (Pg38). The MRV system already exists: the acoustic counters, the NDVI feed, the soil grid — the same instruments that steer the mist also certify the credits. Not counted in the FX bridge (nature credits are not FX-grade exports): this is the program's second ledger, and the scoreboard publishes it the same way.

## The credit spectrum — every financial credit the country earns

| CLASS                            | WHAT IT IS                                                                                                               | THE HONEST NUMBER                                                                                | STATUS                 |
|----------------------------------|--------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------|------------------------|
| Carbon credits                   | 280 kt CO <sub>2</sub> /yr avoided + mineralized + sequestered (solar displacement, the board, the restored belt's soil) | \$1.4-4.2M/yr at \$5-15/t — the compliance/voluntary band; VERIFIABLE, Phase 0 baseline opens it | creditable             |
| Biodiversity credits             | the restored corridor belt, certified by the same sensors that steer the mist                                            | \$0.03-0.25M/yr at the nascent low band                                                          | creditable, nascent    |
| Debt-for-nature swap             | \$500M of sovereign debt refinanced against measured nature outcomes                                                     | ~\$7.5M/yr coupon savings (Gabon/Ecuador precedent, Pg38)                                        | instrument, negotiable |
| ESG-linked debt discount         | the health KPIs reprice the program's own debt                                                                           | 100-150bp on the program portfolio — already counted in the water stack                          | booked                 |
| Concessional windows             | GCF, GEF, Adaptation Fund, EU Global Gateway co-financing — the program's public KPIs are the eligibility                | blended-finance tranches on solar/desal/health — applied, not assumed                            | grant/soft-loan        |
| Rating-linked spread compression | B- to BB- sovereign re-rating on the measured path                                                                       | the state's upside of \$1.2-2.0B/yr on the refinancing share — NOT booked in the program ledger  | the state's harvest    |

the credit spectrum is the program's third ledger — carbon, biodiversity, the swap, concessional windows — each class bookable, none double-counted against the FX bridge, each with its MRV already instrumented. The country's nature, measured by the country's own sensors, becomes the country's credit line.

## The credit rating, projected — the way the agencies rate

| FACTOR                       | TODAY                                          | AT FULL PHASE                                                                                                                       |
|------------------------------|------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|
| Reserves / import cover      | 4.5 months today (measured)                    | FX accumulation at full phase: ~\$1.0-1.4B over the 5-year build → 6+ months — the S&P reserves-adequacy factor moves               |
| External position            | CA deficit ~-3.3% of GDP                       | the loop narrows it by ~0.5 pts (279M/yr) → ~-2.5-2.7% — the agencies rate the PATH, not just the level                             |
| Growth                       | ~1.9% today                                    | +0.5 pts from the loop + 1.2 pts from the peace dividend → 3-4% — the strongest single upgrade factor for B-rated sovereigns (Pg40) |
| Fiscal                       | subsidy burden + debt service 26.2% of exports | the substitution legs + the swap's coupon savings + the ESG delta relieve the budget on the same lines                              |
| Institutional predictability | policy volatility is the chronic negative      | the public scoreboard, the gates, the apolitical program — a measurable governance signal the methodology rewards                   |

**The path:** B- → **BB- / Ba3 within 3-5 years** (reserves 6+ months, CA ≤ -2.5%, growth ≥ 3% sustained — all inside the measured numbers) and **BB / Ba2 in 5-7 years** with the swap and the nature ledger. Investment grade is NOT claimed: that requires debt/GDP ≤ 60% and a decade of institutional track record — the honest frontier.

the spread arithmetic, stated for what it is: B- to BB- sovereign compression is ~300-500bp; on the refinancing share of the \$40.5B stock that is the state's upside of \$1.2-2.0B/yr — NOT booked in the program's ledger: only the swap's measured \$7.5M/yr and the program-debt ESG delta are credited. The rating is the state's harvest; the program only plants.

## Air quality — the air the children breathe

The board plant caps the phosphogypsum stack dust; the solar leg displaces the grid's gas burn; the biocontrol corridor replaces pesticide drift; and the dew the mist buys scavenges PM from the air (wet deposition). Measured by the PM2.5/NOx/SO2 sensor grid, the dispersion model and the cohort's respiratory KPIs — air quality enters the scoreboard the same way the water does.

## The last inventory — nothing forgotten, nothing hidden

| ITEM                                 | THE GAP IT CLOSES                                                                            | ITS HOME                                                                                                                                                                                                 | STATUS              |
|--------------------------------------|----------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|
| Pond blowdown (the salt bleed)       | the spirulina ponds' own concentrate, when the recycled medium's salts saturate              | routed to the brine line — the pond's salt joins the valorization stream; the water loop closes, the salt loop closes                                                                                    | routing row, added  |
| Construction carbon & energy payback | the embodied footprint of building the park (PV, concrete, steel) vs the operating recovery  | PV energy payback ~1-2 years (industry standard, DERIVED); the park's embodied carbon is repaid by the displacement + mineralization inside the first operating years — computed in the Phase 0 LCA item | LCA item in Phase 0 |
| The risk-transfer layer              | drought years, PV underperformance, price shocks — the hazards the gates alone cannot absorb | parametric insurance on the mist corridor's target (rainfall index) and the solar PPA (irradiation index); the degraded cascade already computed is the self-insurance floor                             | insurance, bookable |

| ITEM                      | THE GAP IT CLOSES                                                                                   | ITS HOME                                                                                                                                                                                                   | STATUS                             |
|---------------------------|-----------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------|
| Nuclear export compliance | U&sb3;O&sb8; leaves the country — the chain of custody the IAEA/NPT regime demands                  | the CNSTN gate already named extends to export controls: buyer compliance, safeguards accounting, transport certification — the Cameco-class compliance chain, cited as a gate, not an afterthought        | compliance chain, gate             |
| The women's leg           | 54% of women graduates unemployed — who exactly gets the jobs must be designed, not hoped           | the nursery, the workshop, the pond crews and the health workers as women-led cooperatives in the blockade towns — the quota principle already proven in the terfas domestic clause, applied to employment | cooperative structure, contractual |
| The AI's own audit        | the dispatcher's code and the twin's models must be auditable or the scoreboard's trust is borrowed | open models, versioned, the prediction-error KPI published, independent code audit before each phase — the nervous system is inspected like everything else it measures                                    | code audit, gate                   |

## The fauna water ladder — the earwig's drop, as the unit of the whole scale

| SPECIES              | MASS   | WATER / DAY                  | THE NOTE                                                                                                           |
|----------------------|--------|------------------------------|--------------------------------------------------------------------------------------------------------------------|
| The earwig           | 0.1 g  | ~0.5 µL/day (measured class) | one 50 µL drop = 100 earwig-days — the owner's own observation, as the unit of the whole scale                     |
| The wild bee         | 0.1 g  | ~30 µL/day (measured)        | a 10 mL/m <sup>2</sup> dew morning serves ~300 bees per square meter                                               |
| The desert lizard    | 20 g   | ~1.4 mL/day (FMR)            | dew-licking Acanthodactylus — the Saharan micro-fauna's classic water behavior                                     |
| The sandgrouse       | 250 g  | ~10 mL/day (FMR)             | the Sahara's dew-drinker; chicks absorb water through belly feathers — the corridor's fog mornings are its nursery |
| The fennec fox       | 1.2 kg | ~36 mL/day (FMR)             | the desert's smallest canid — water from prey, dew and the corridor's troughs                                      |
| The Egyptian vulture | 2 kg   | ~54 mL/day (FMR)             | globally endangered, breeds in Tunisia — the corridor's carrion-cycling scavenger                                  |
| The Dorcas gazelle   | 17 kg  | ~298 mL/day (FMR)            | endangered; its historic southern steppe is the restoration belt itself                                            |
| The Barbary sheep    | 50 kg  | ~706 mL/day (FMR)            | the mountains' own ungulate — the corridor's upslope fringe is its range                                           |
| The loop's flock     | 60 kg  | ~800 mL/day                  | the agrivoltaic sheep: the mist's 2% is its drinking trough, and its grazing is the land's firebreak               |

one average day's dew on the corridor — 51 m<sup>3</sup>, the same water as one house's annual use — rations across the whole ladder: 10<sup>11</sup> insect-days or 1.7 billion bee-days or a million vulture-days or 170,000 gazelle-days. The species that drink it are the ones Tunisia is losing: the scale that ends at the Dorcas gazelle begins at the

earwig's half-microliter — and the mist corridor is the trough for every rung. Allometric spine: daily water =  $0.123 \times \text{mass}^{0.8}$  mL (Nagy, Pg43), cross-checked against measured insect rates.

**The spirulina health claim, at the evidence:** and the health claim, downgraded to the evidence: the 2025 meta-analysis shows positive weight/MUAC effects in malnourished children (Pg41); the 2026 meta-analysis finds NO significant growth effect (Pg42). The page claims the FOOD, not the cognition — the cohort measures the rest, and the gates decide.

**What this document is:** a concept-level engineering design whose numbers are computed in code from sourced primitives — recomputed, corrected, and applied into the design. Free and public.

**What it is not:** a commissioned state program, a bank-grade feasibility study, an investment offer, or a claim that any of this has been built. Nothing here is advice to any government or any investor. Nothing replaces geological, radiological, geotechnical or EPC-level surveys. Verify every figure and citation before relying on any of it.

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